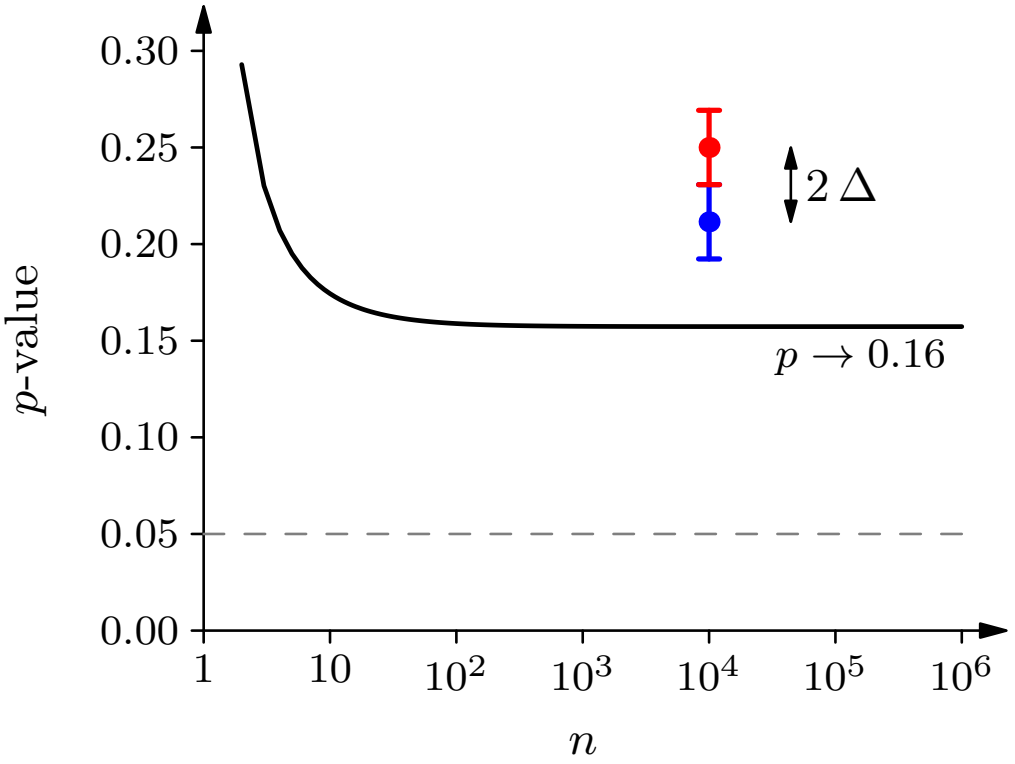
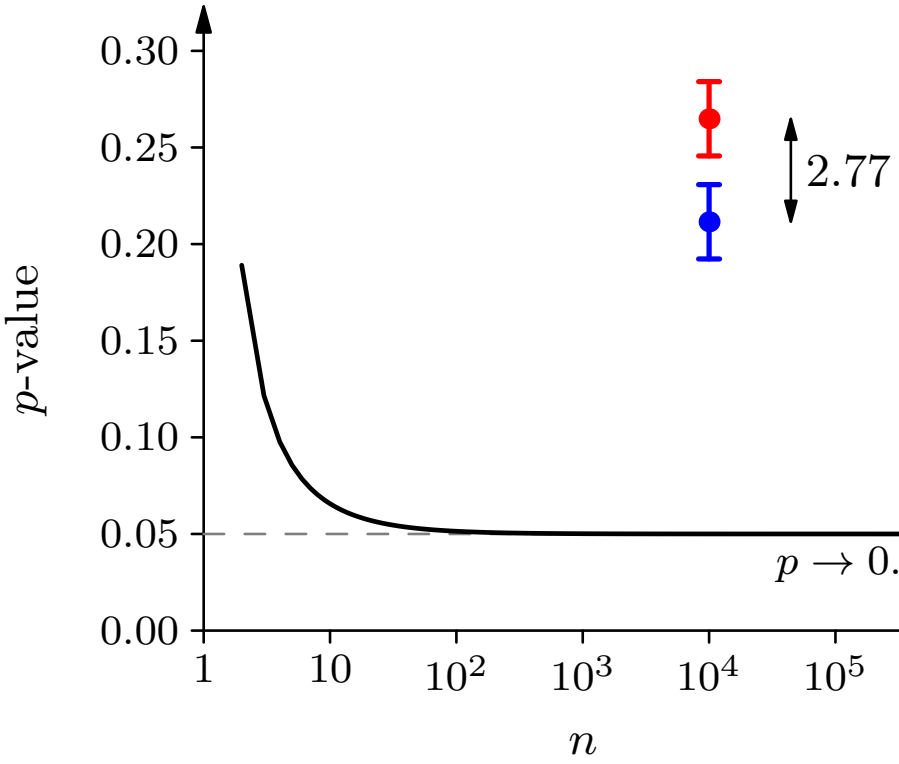
Mathematics of Machine Learning

On Becoming a Scientist

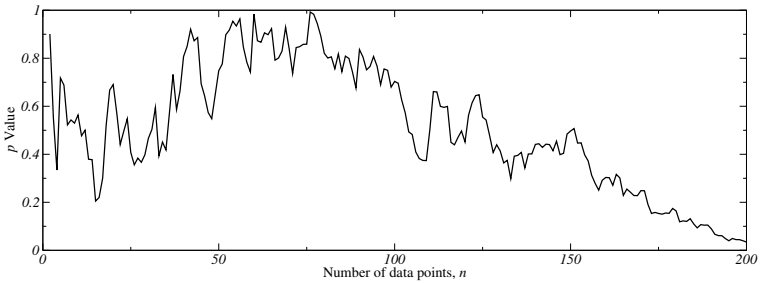




interpreting data, standard errors, confidence tests

Outline

1. **Reporting Experiments**
2. Experimental Error
3. Confidence Estimates



Experiments and Reproducibility

- One of the fundamental requirements of science is that experiments can be reproduced
- This applies to machine learning and deep-learning as it does to particle physics, molecular biology, etc.
- We need to describe completely what procedures were carried out
- However, experiments have natural variability so what do we mean by repeatability?

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Describing Your Experiments

- You need to tell the reader
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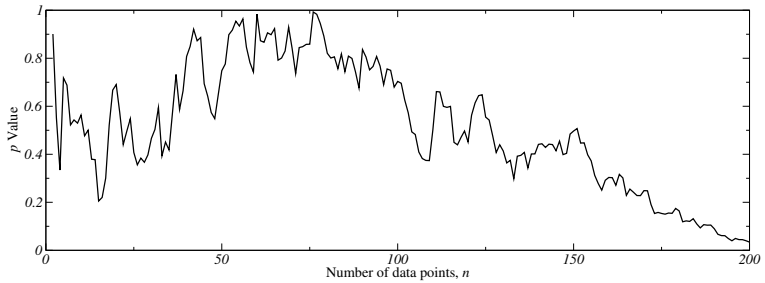
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- Any empirical result is useless without some estimate of error
- If you tell me algorithm A has a performance of 95% and algorithm B has a performance of 97%, how do I know that this didn't happen by chance?
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- We assume that our test set consists of typical examples that we might encounter and that they are independent (e.g. we are not seeing the same example over and over again)
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- We need to think about it and if they are not true you may need to collect new data or be very cautious of the claims that you are making
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Collecting Statistics

- Given some experimental measurements $\{X_i | i = 1,2,\ldots,n\}$ we can estimate the mean and variance

$$\hat{\mu}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \qquad \hat{\sigma}_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \hat{\mu}_n)^2$$

- These are not the true mean, $\mu = \mathbb{E}[X_i]$, and true variance $\sigma^2 = \mathbb{E}\left[(X_i - \mu)^2\right]$, but they are unbiased estimates in that

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- Consider the mean squared error (MSE) of the estimator $\hat{\mu}_n$ (independent X_i)

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- Traditionally we would write an experimental result as $\hat{\mu}_n \pm \Delta$

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- Not 1.6582356 ± 0.122543 but 1.7 ± 0.1

Repeated Bernoulli Trials

- Suppose we are measuring a binary outcome, success or failure
- Assuming the trials are independent (so called Bernoulli trials) then the only unknown is the probability, p, of a success
- If we make n trials then probability of k successes is given by the binomial distribution

$$\mathbb{P}(K = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

- The mean and variance of K are

$$\mathbb{E}[K] = np \qquad \mathbb{E}\left[(K - np)^2\right] = np(1-p)$$

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Repeated Bernoulli Trials

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$$\hat{p} = \frac{K}{n}$$

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- The estimated error in the mean is given by the root mean squared error (RMSE) $\sqrt{\mathbb{E}[(\hat{p}-p)^2]} = \sqrt{\frac{p(1-p)}{n}}$
- We don't know p, but we can use an estimate, $\hat{p} = \frac{K}{n}$

$$\Delta \approx \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

- We can also use a worst-case bound (when $p = \frac{1}{2}$)

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Errorbars

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- Traditionally errorbars show the estimated error in the mean (standard error, $\Delta = \hat{\sigma}_n/\sqrt{n}$)
- For roughly normal data the true mean is outside the errorbar about 32% of the time
- Some people use 95% confidence intervals as errorbars, saying 95% of the time the true mean lies within the errorbars
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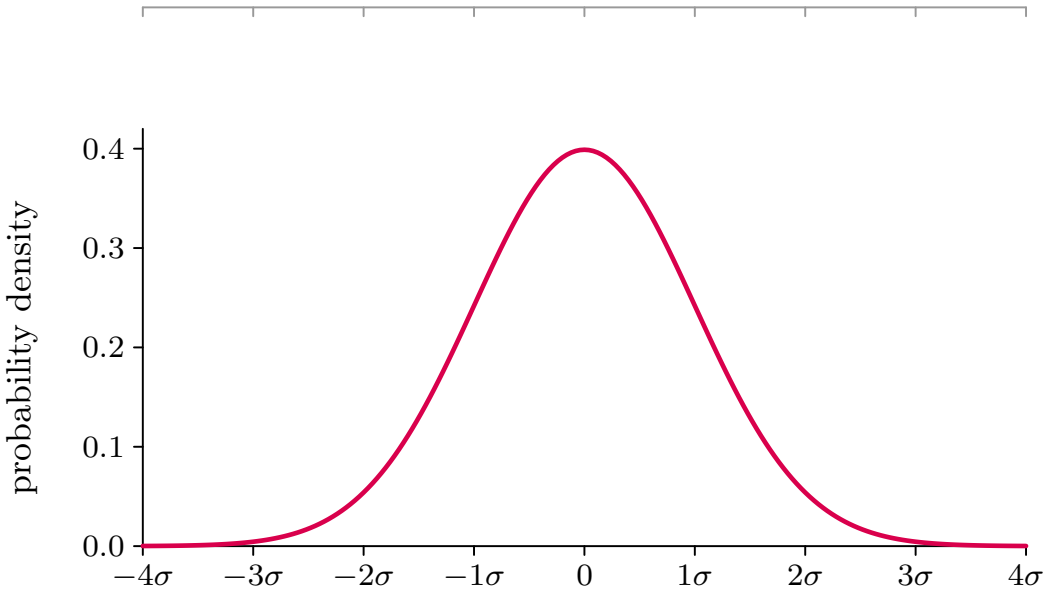
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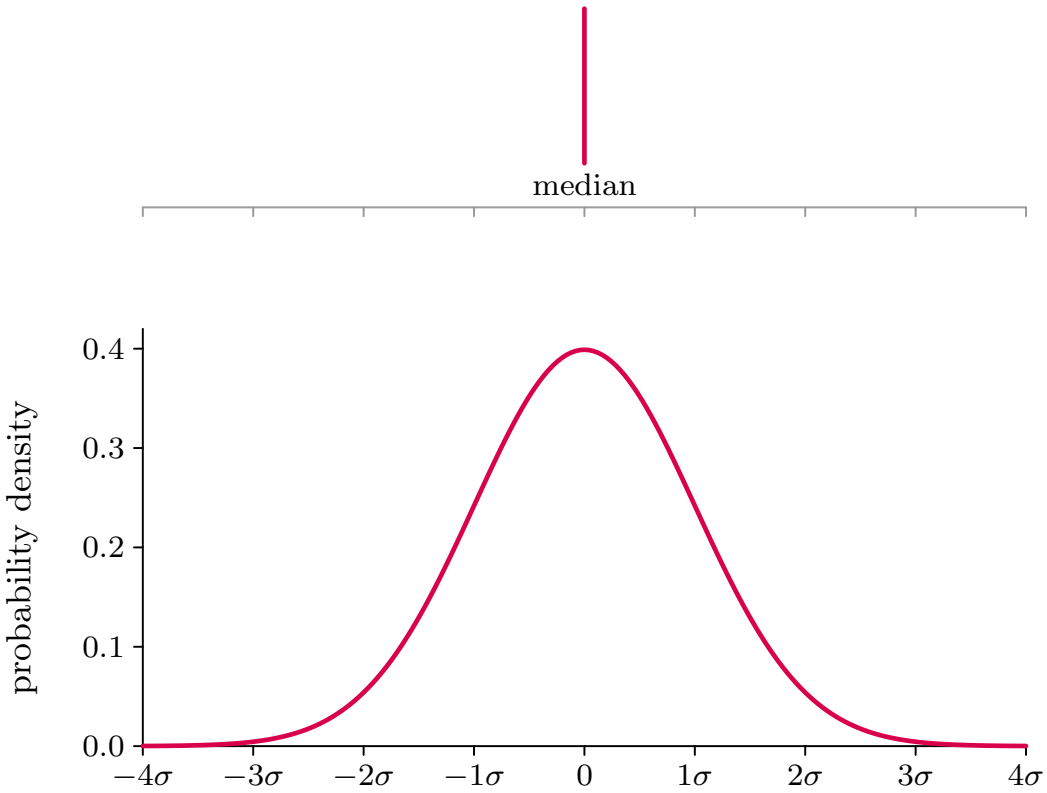
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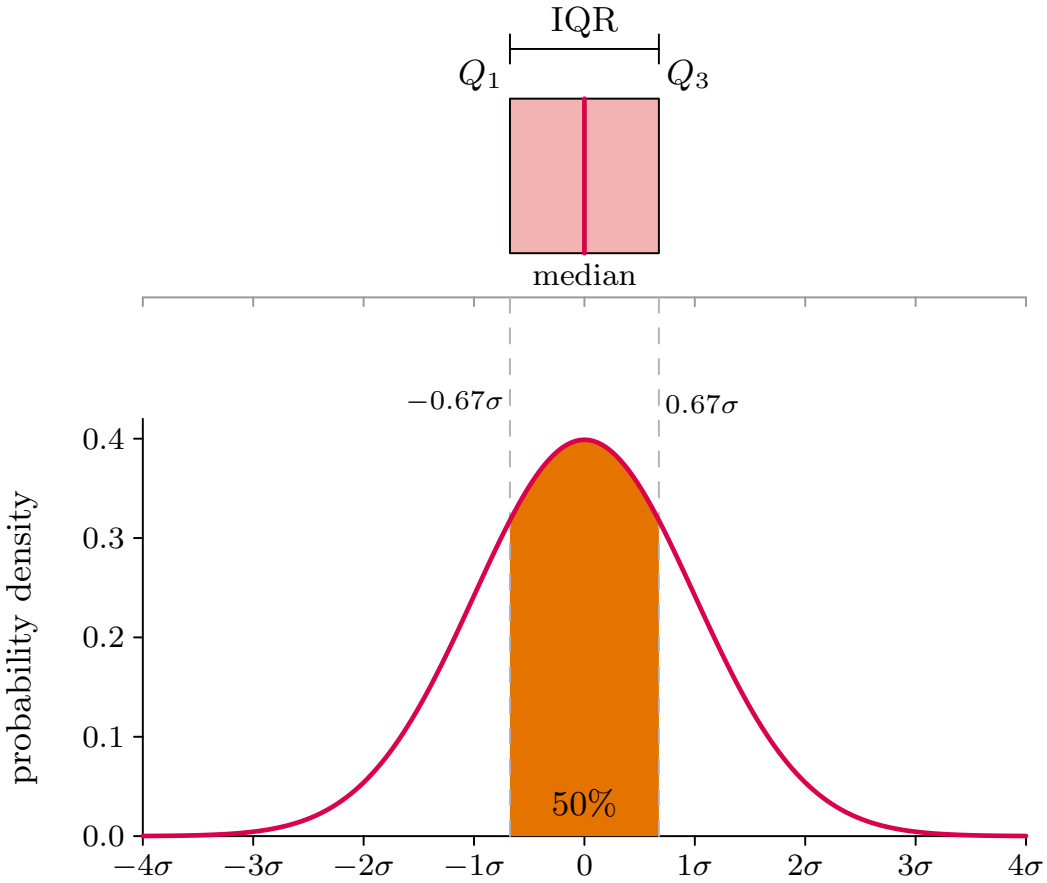
Box Plots


probability density
0.4
0.3
0.2
0.1
0.0
-4σ
-3σ
-2σ
-1σ
0
1σ
2σ
3σ
4σ

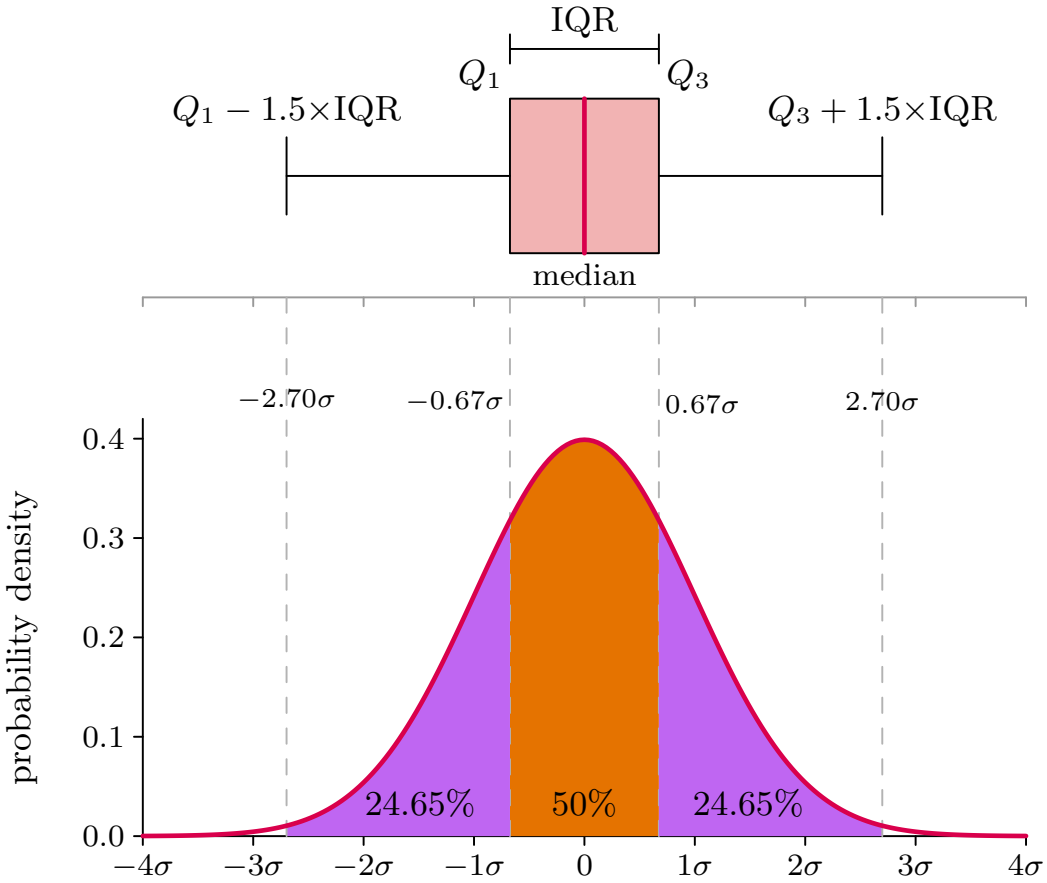
Box Plots


median
probability density
0.4
0.3
0.2
0.1
0.0
−4σ
−3σ
−2σ
−1σ
0
1σ
2σ
3σ
4σ

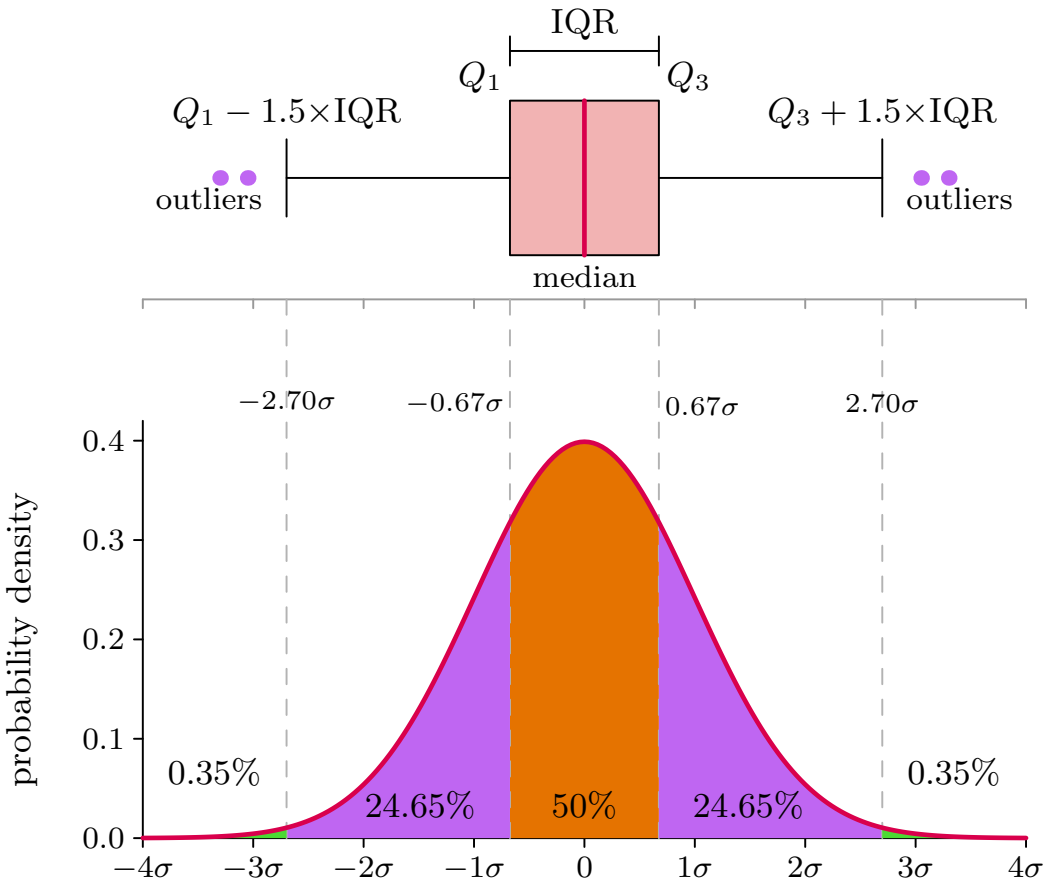
Box Plots


IQR
Q_1
Q_3
median
-0.67σ
0.67σ
probability density
0.4
0.3
0.2
0.1
0.0
50%
-4σ
-3σ
-2σ
-1σ
0
1σ
2σ
3σ
4σ

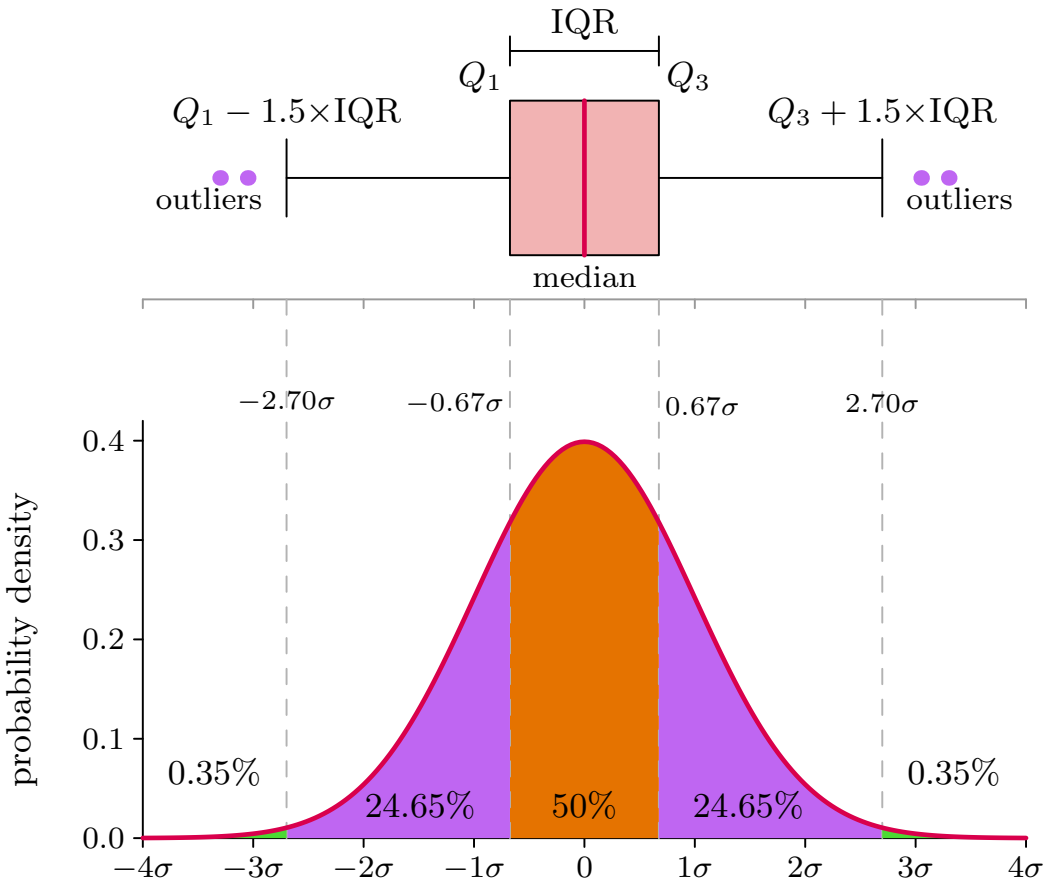
Box Plots


IQR
Q_1
Q_3
$Q_1 - 1.5\times\text{IQR}$
$Q_3 + 1.5\times\text{IQR}$
median
-2.70σ
-0.67σ
0.67σ
2.70σ
probability density
0.4
0.3
0.2
0.1
0.0
24.65%
50%
24.65%
-4σ
-3σ
-2σ
-1σ
0
1σ
2σ
3σ
4σ

Box Plots


IQR
Q_1
Q_3
$Q_1 - 1.5\times\text{IQR}$
$Q_3 + 1.5\times\text{IQR}$
outliers
outliers
median
-2.70σ
-0.67σ
0.67σ
2.70σ
probability density
0.4
0.3
0.2
0.1
0.0
0.35%
24.65%
50%
24.65%
0.35%
-4σ
-3σ
-2σ
-1σ
0
1σ
2σ
3σ
4σ

Box Plots

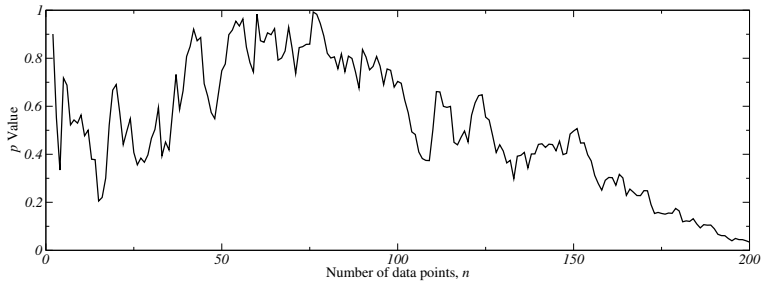


Box plots summarise the *spread* of the data.

They show no error bars on the median, so you cannot tell from them whether two results are statistically significantly different.

Outline

1. Reporting Experiments
2. Experimental Error
3. **Confidence Estimates**

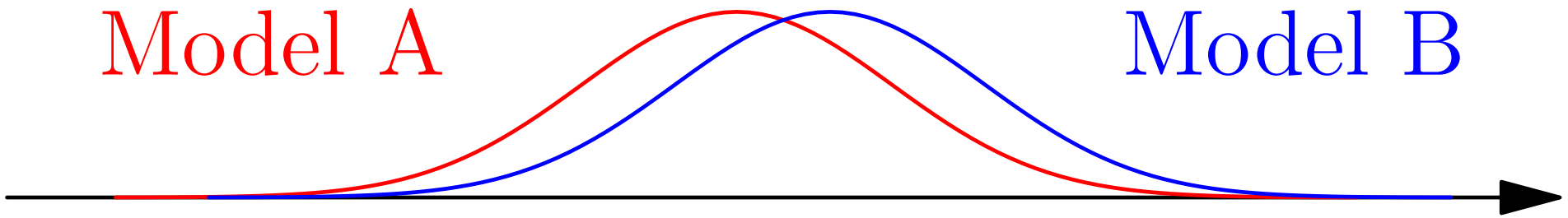


Know What's Best

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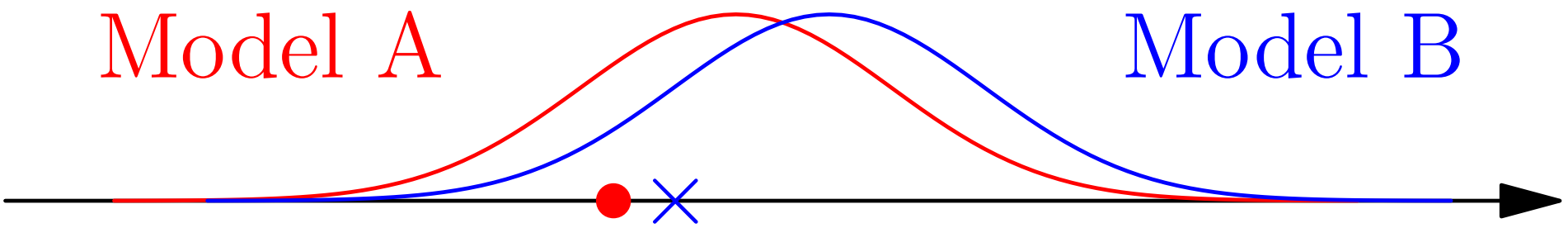
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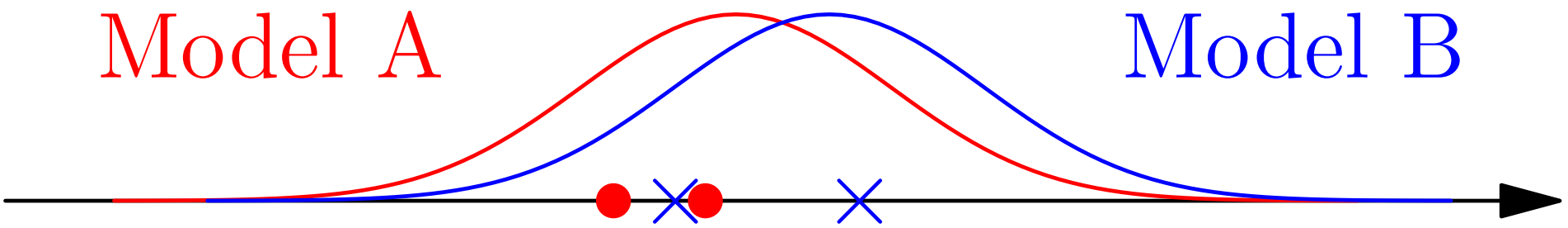
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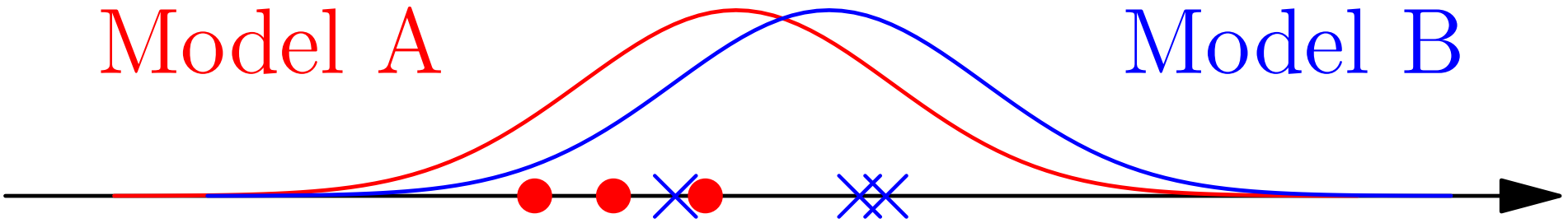
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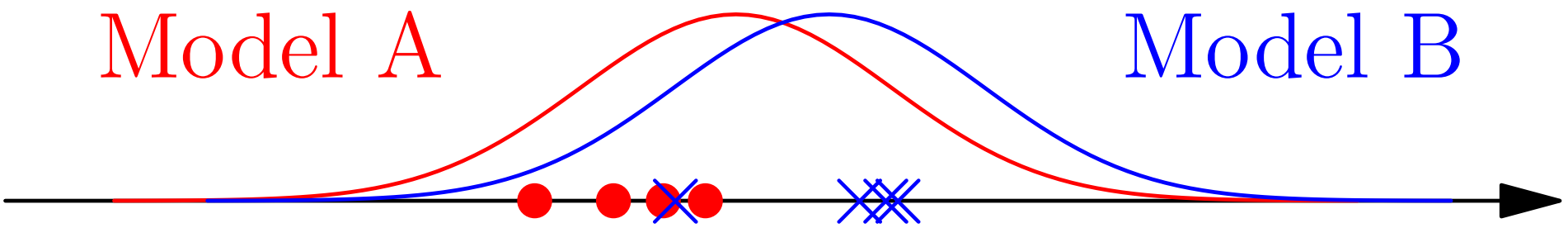
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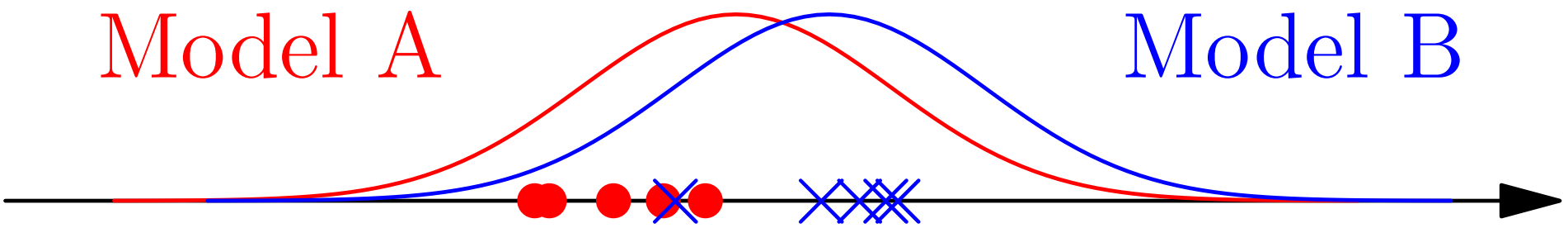
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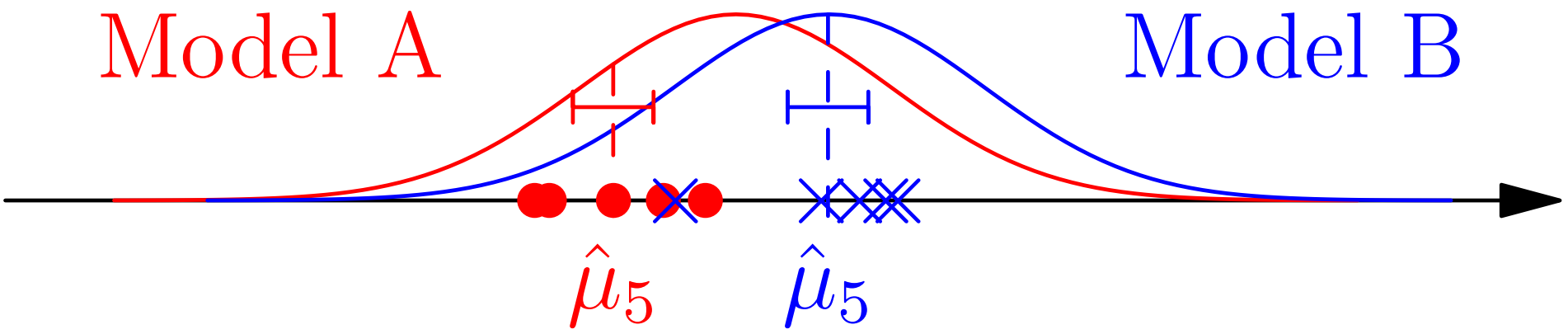
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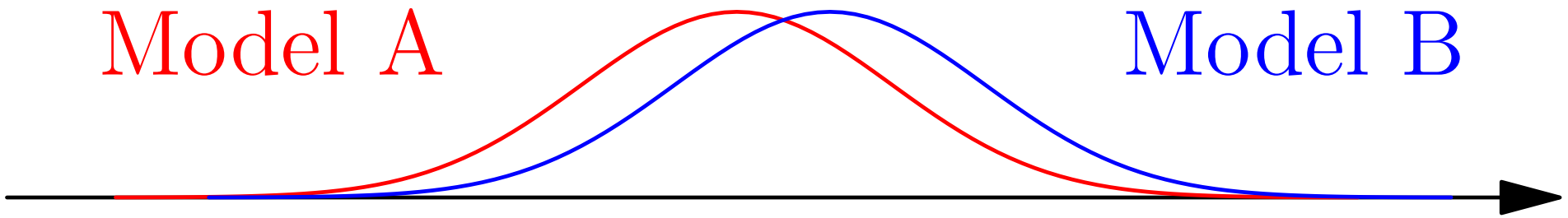
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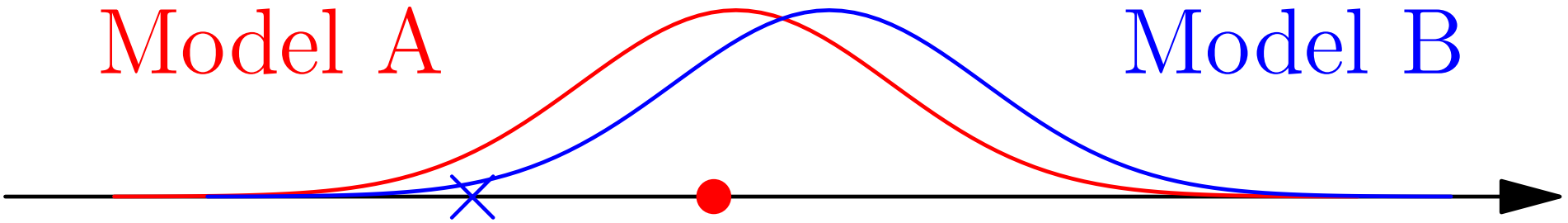
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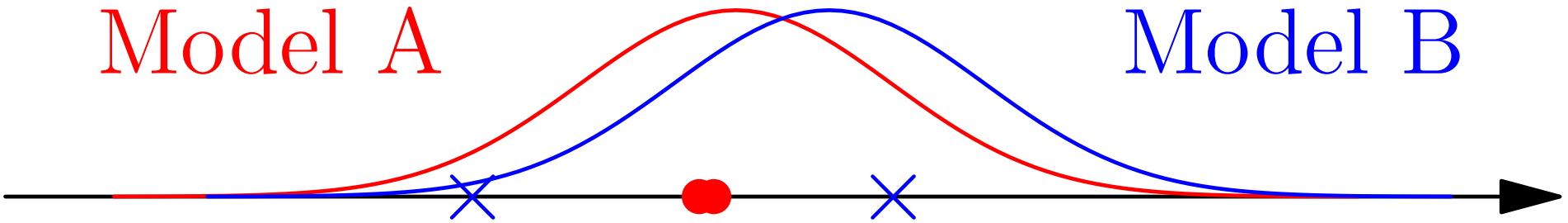
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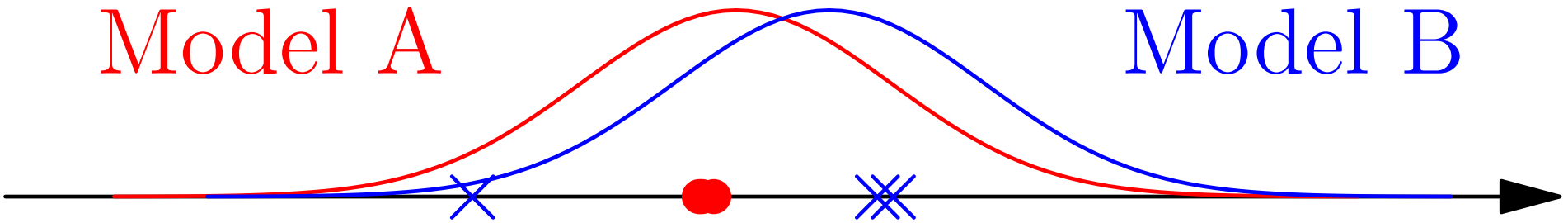
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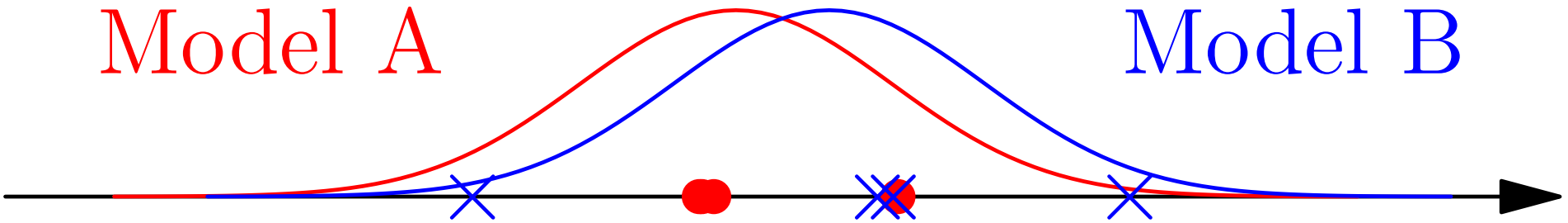
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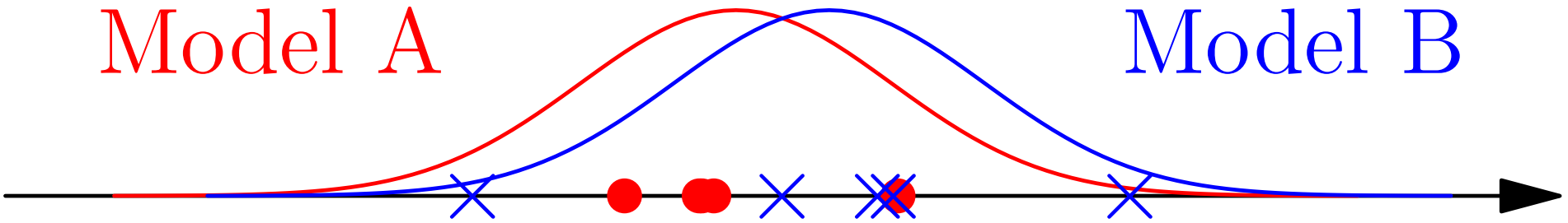
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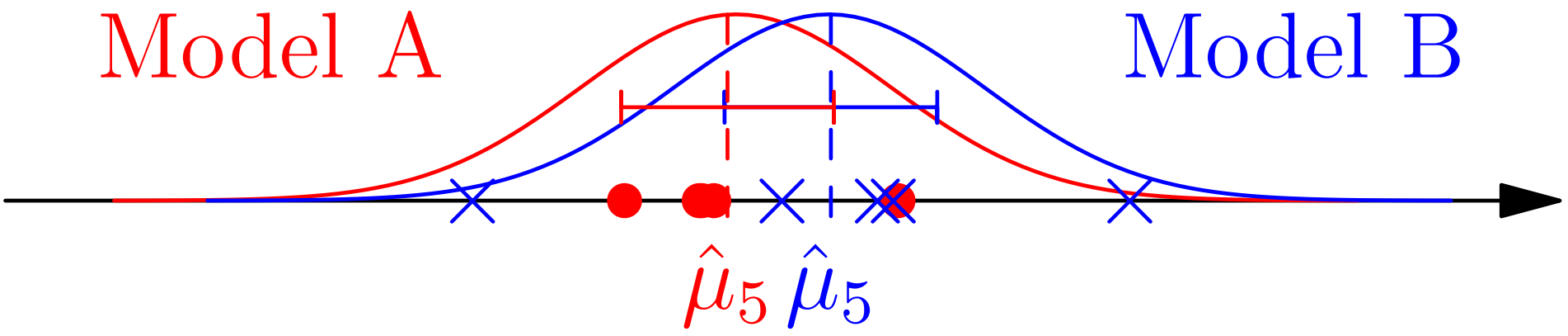
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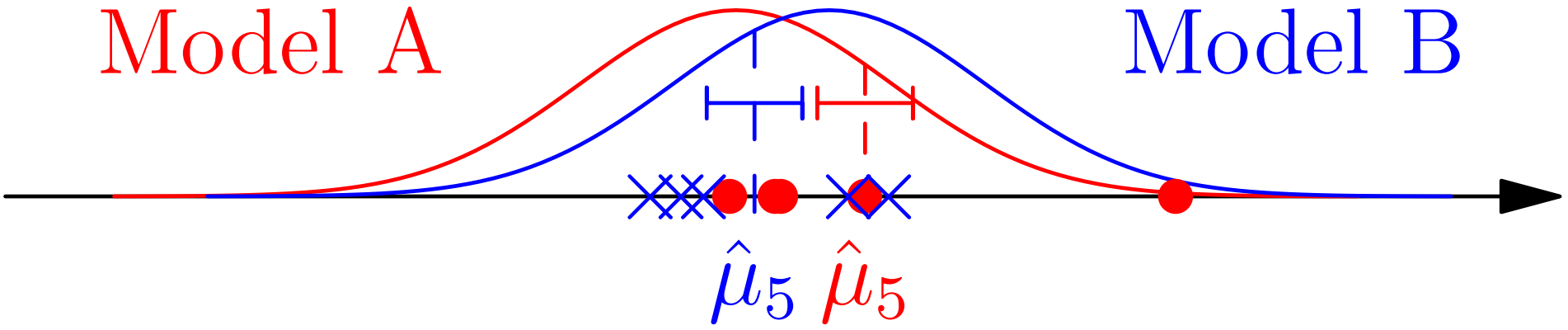
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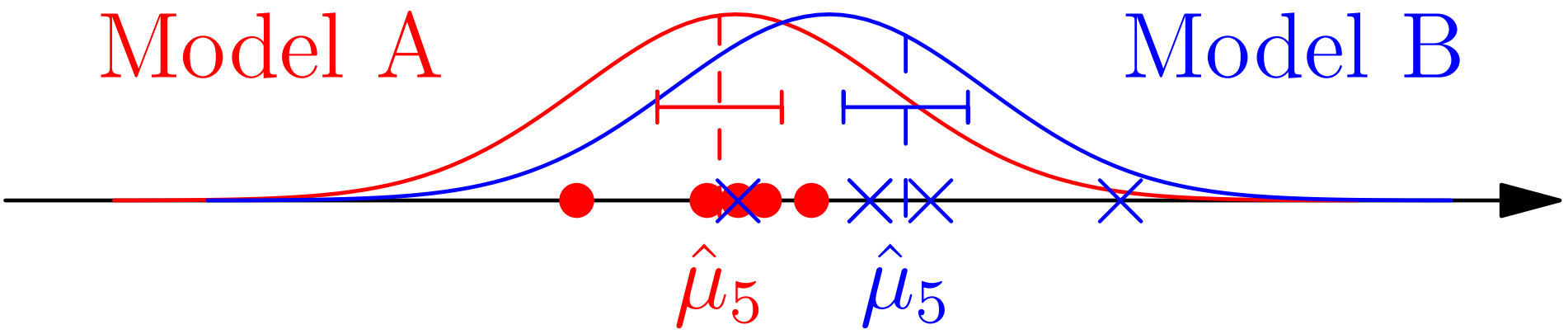
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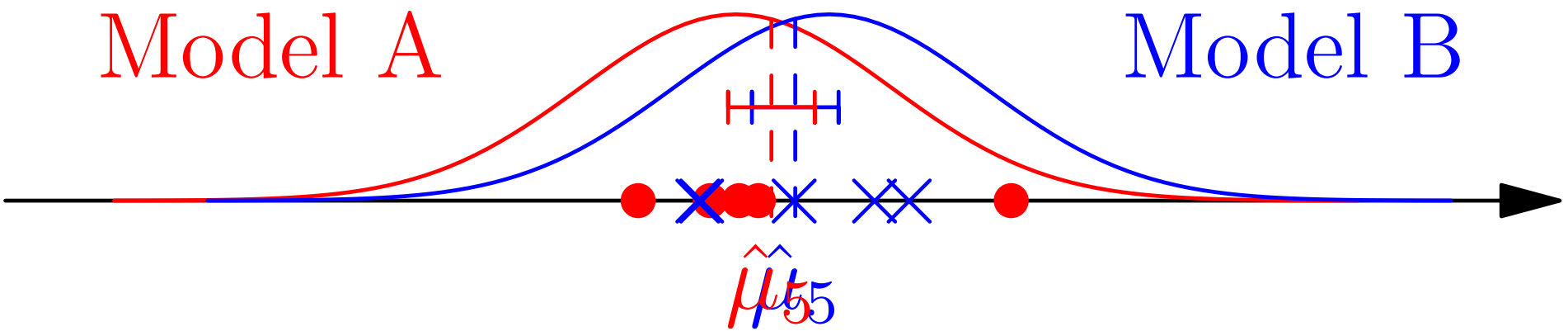
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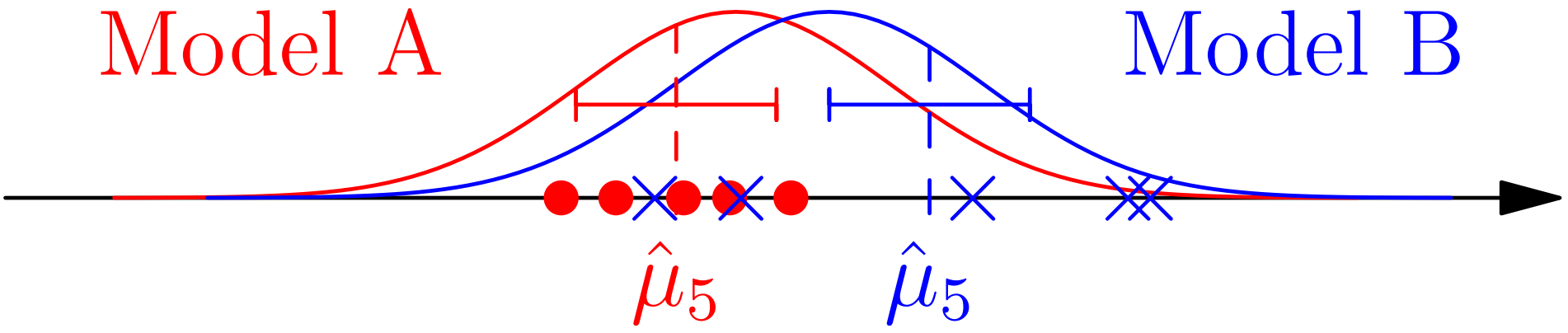
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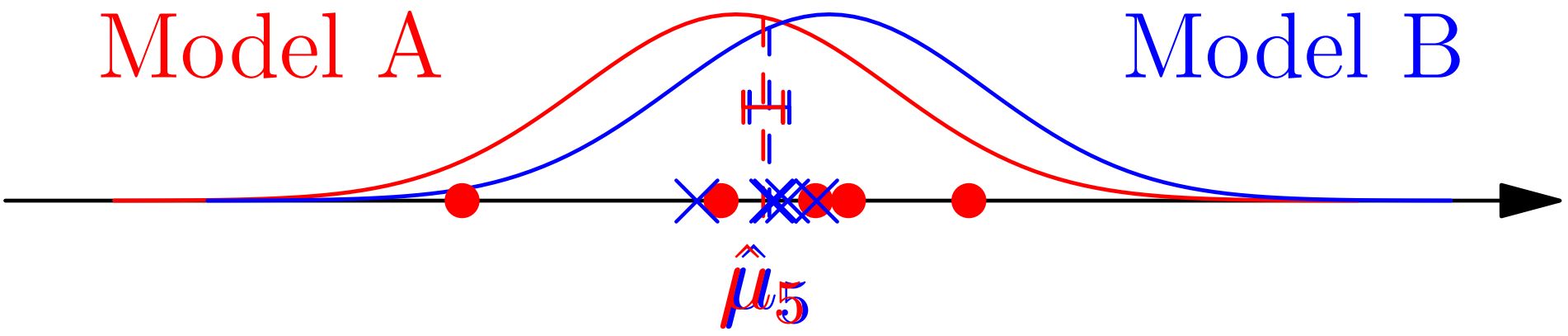
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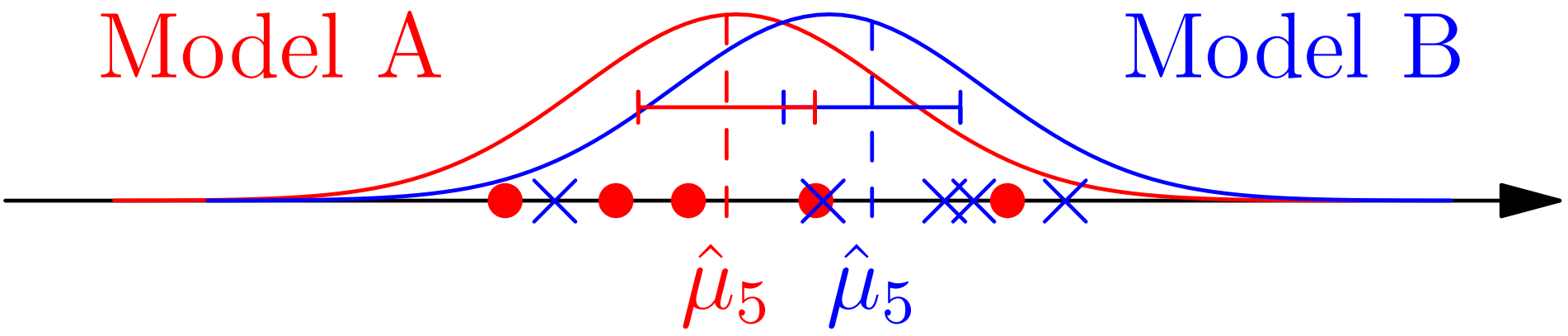
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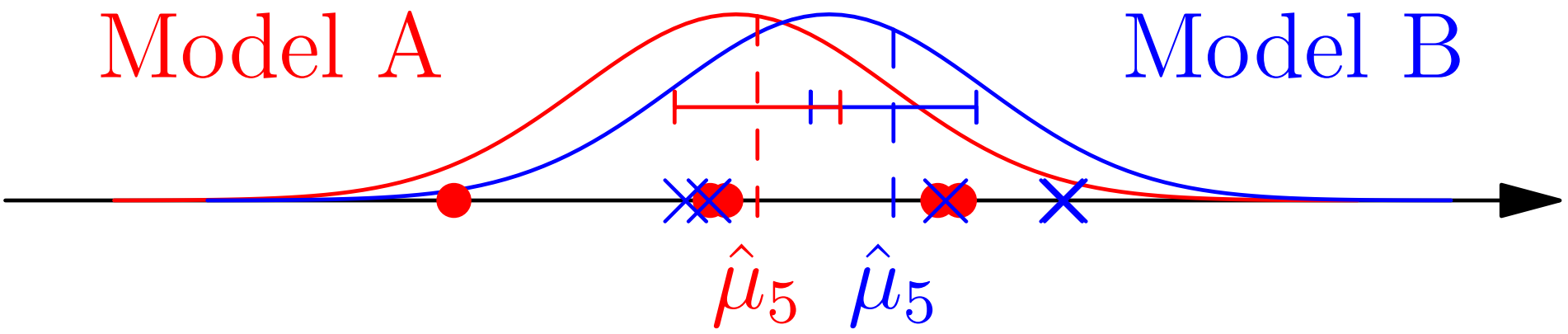
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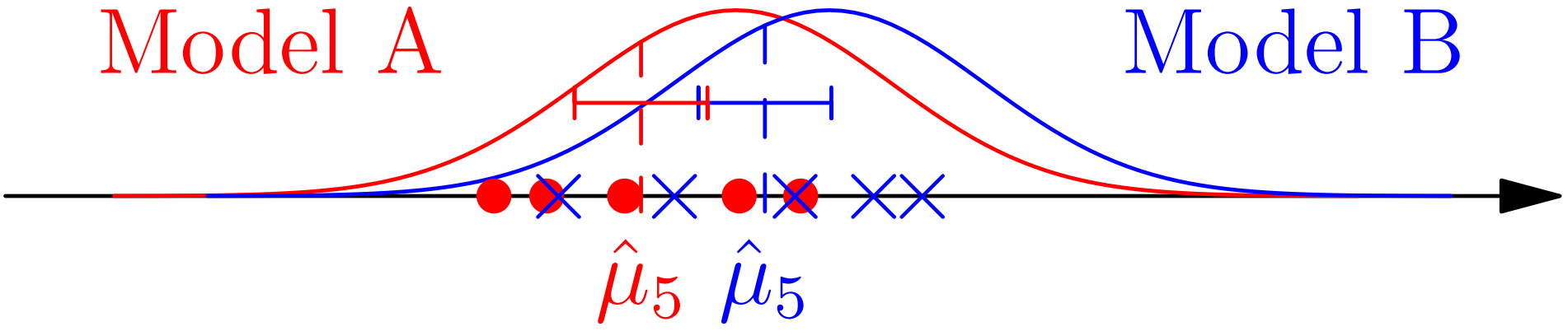
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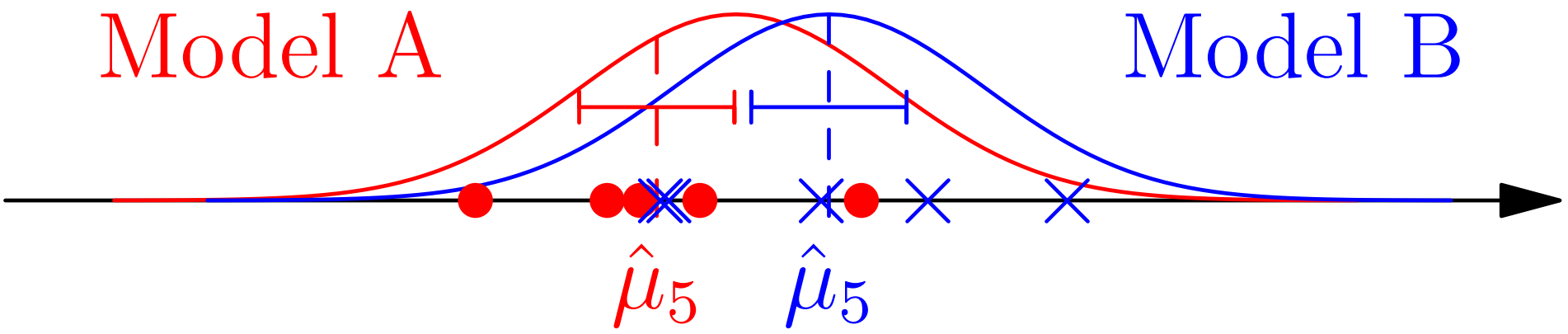
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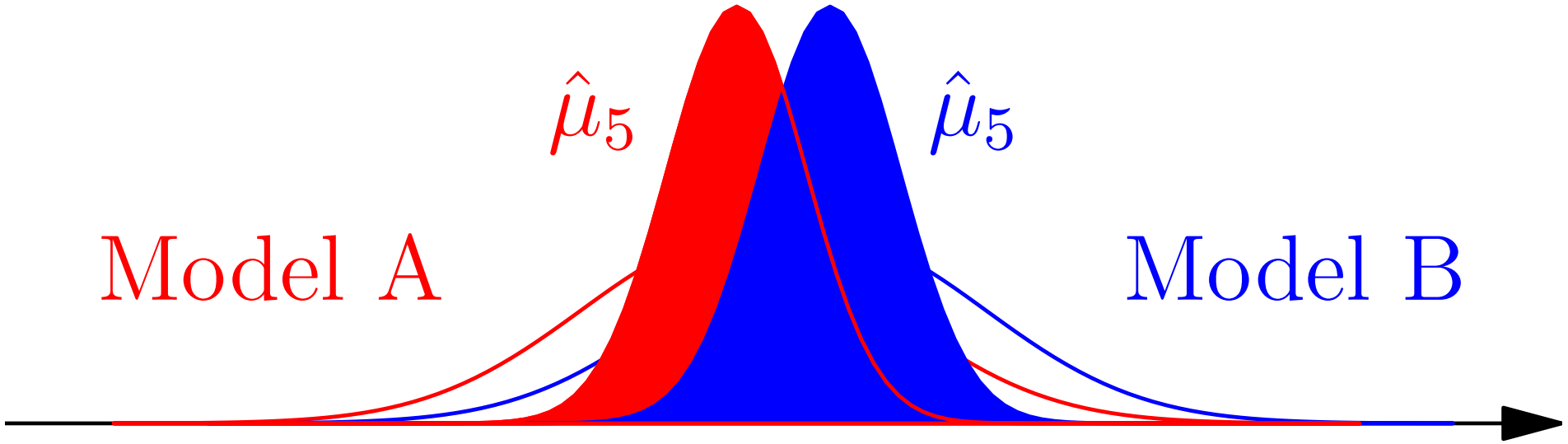
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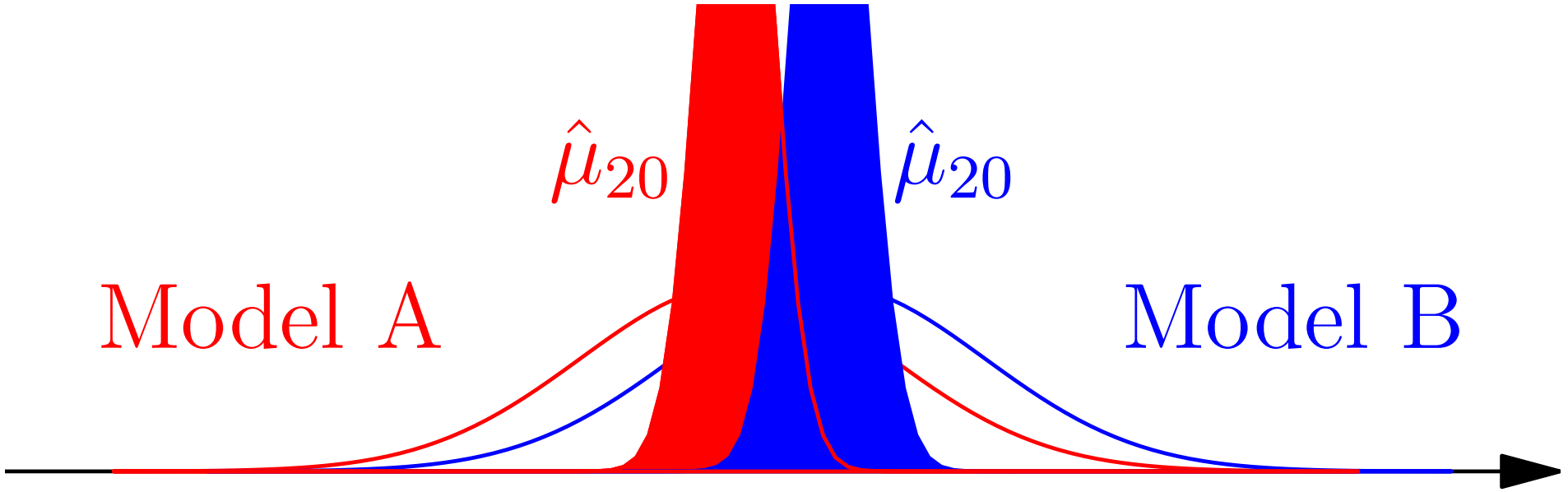
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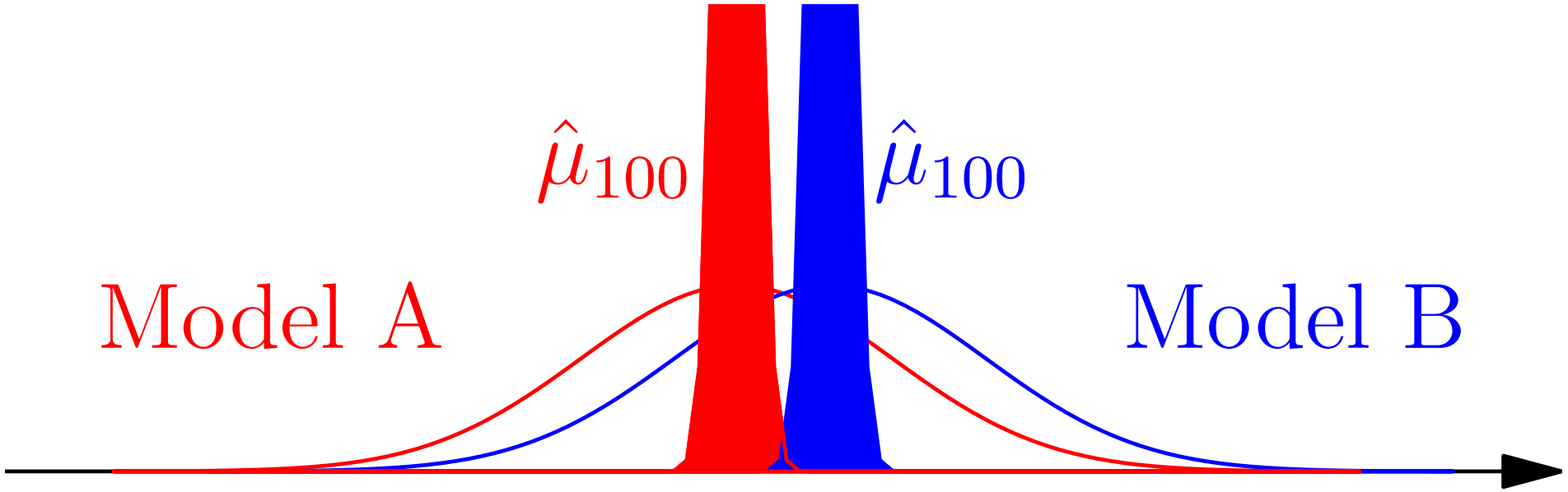
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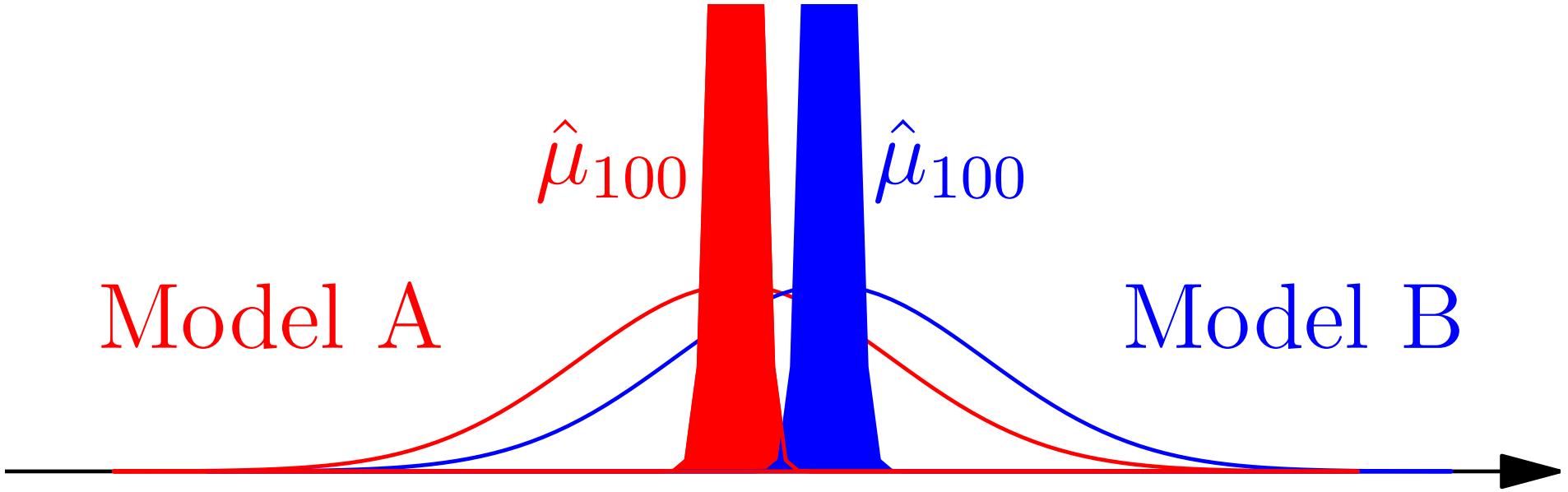
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The more data we have the less likely we will make a mistake

Quantifying Uncertainty

- We want a way of quantifying the chance that model B is better than model A
- To do this we can consider a **null hypothesis**

$$We\ suppose\ that\ model\ A\ and\ B\ have\ the\ same\ performance$$

- We then ask, under the null hypothesis, what is the chance of obtaining a gap in performance at least as big as the one we observed
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T-tests

- The original t-test was invented by William Gosset who worked for the Guinness brewery in Dublin, but published under the pseudonym $Student$
- He assumed A and B are normally distributed with the same standard deviation; the null hypothesis is that their means are equal
- The more common test now is Welch's t-test, which drops the assumption that the standard deviations are equal
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- It is common to say that A is $statistically\ significantly$ better than B if, under the null hypothesis, the probability of a t at least as extreme as the one observed (the p-value) is less than 0.05
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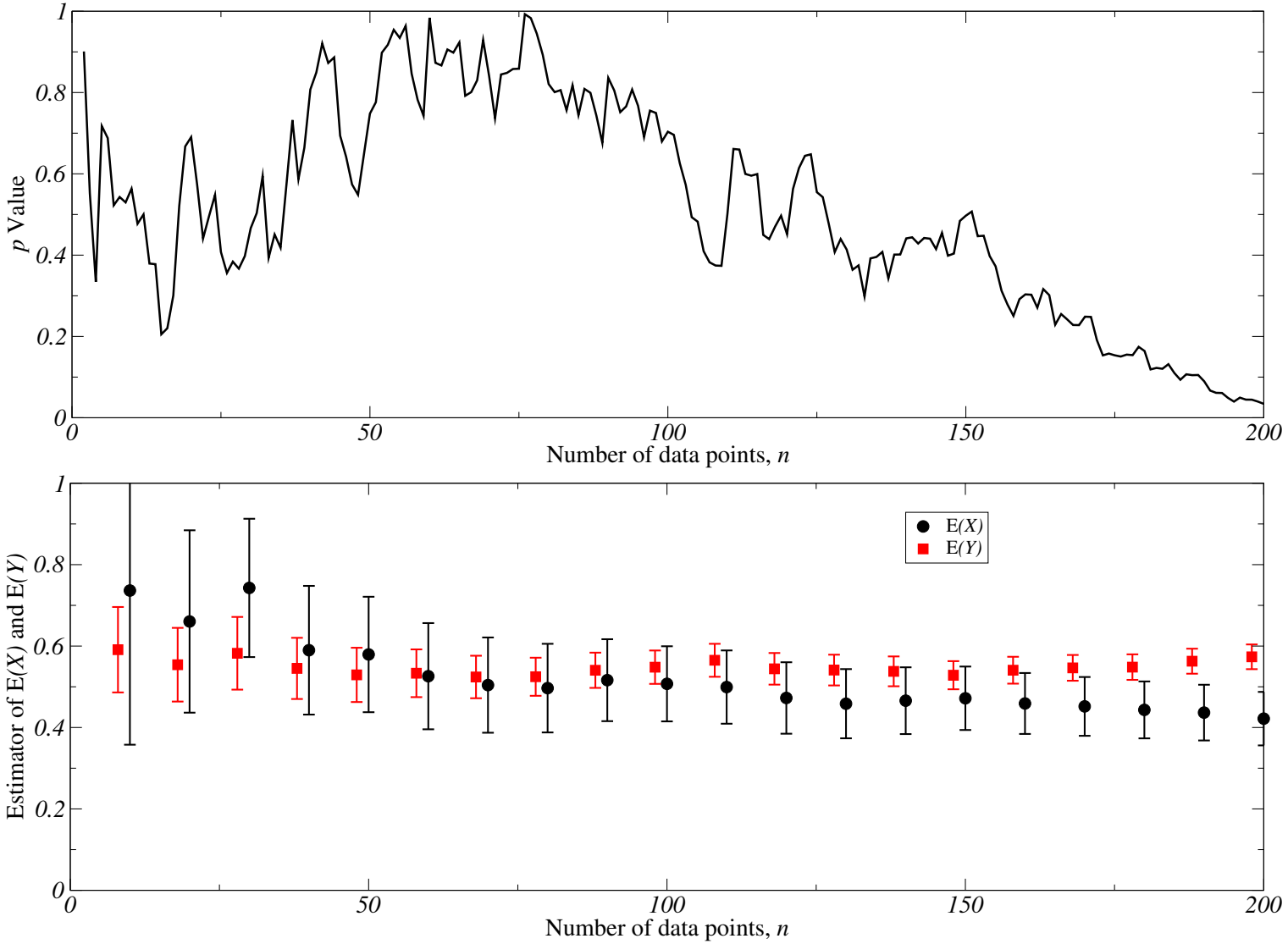
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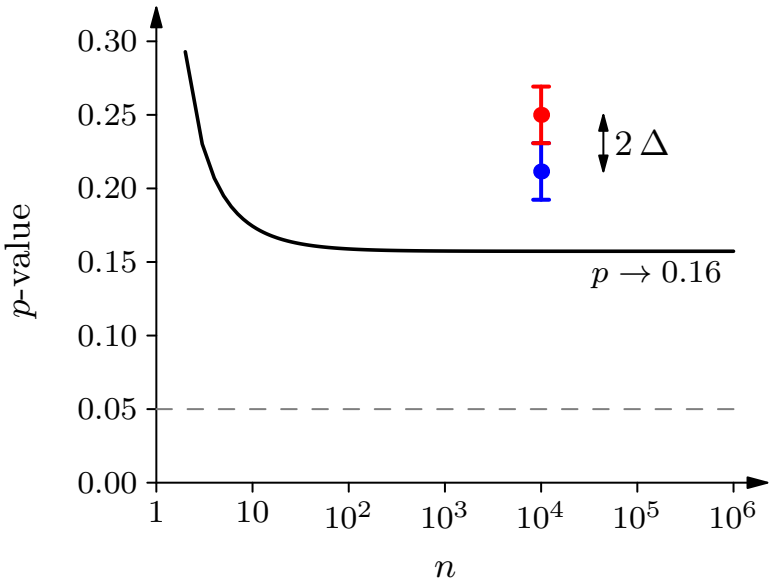
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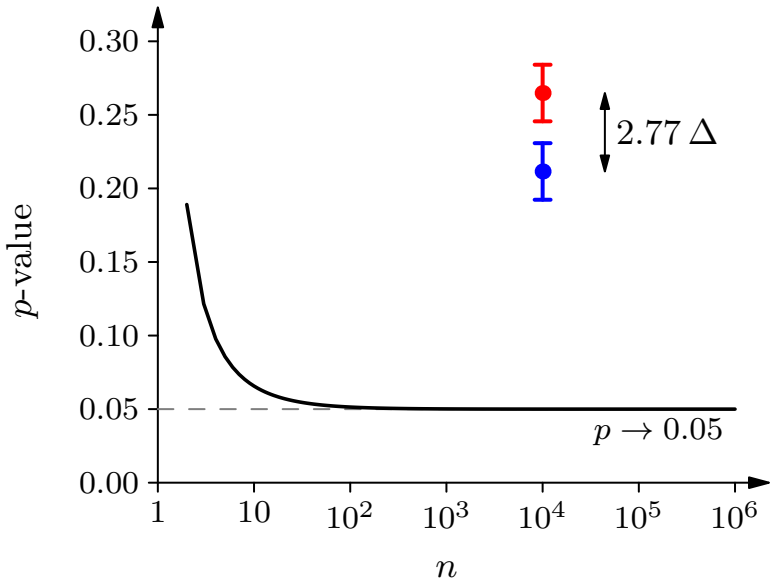
Example

$X \sim \mathcal{N}(0.4,1)$ $Y \sim \mathcal{N}(0.6,0.25)$



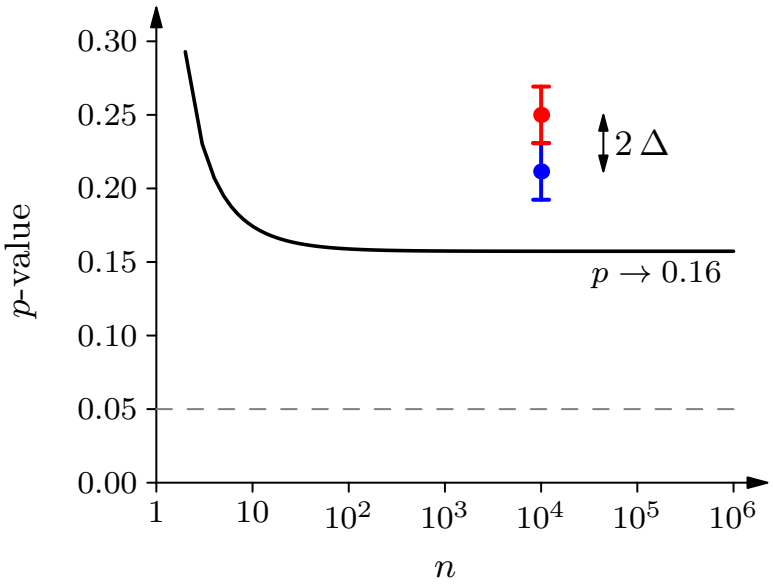
Error Bars and Significance

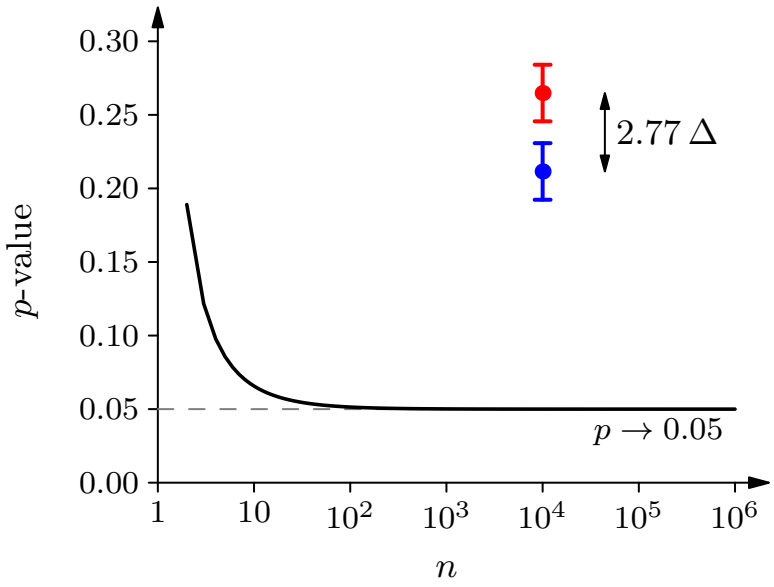




- Separate errorbars (1 standard error) are **not** enough: if they just touch, $p \approx 0.16$
- For $p < 0.05$ the gap must be about 2.8 standard errors

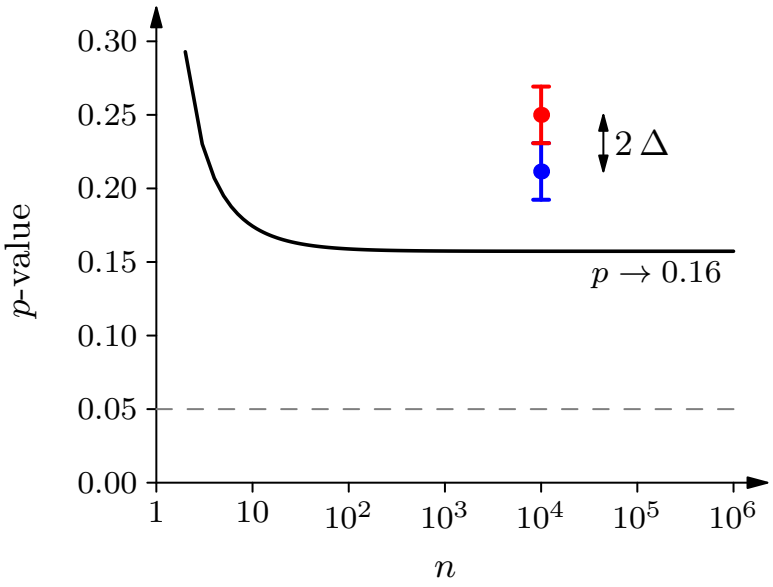
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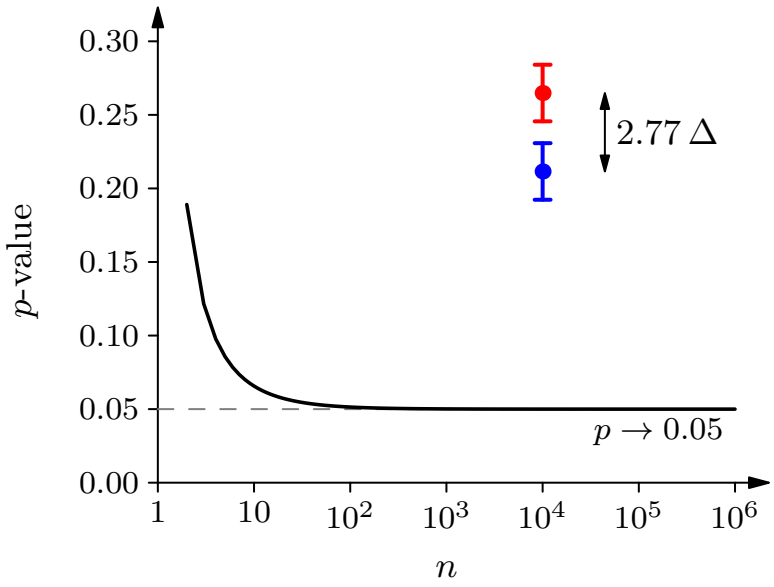




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Paired T-Test

- Sometimes the fluctuations in performance will come both from the test example and from the difference in performance
- For example model A might always be better than model B, but the performance fluctuates depending on the example
- A normal t-test might not show any significant difference
- A paired t-test uses the per-example differences $X_i - Y_i$, whose standard error is

$$\hat{\sigma}^2_{X-Y} = \frac{\hat{\sigma}^2_X + \hat{\sigma}^2_Y - 2\hat{c}_{X,Y}}{n}$$

$$\hat{c}_{X,Y} = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \hat{\mu}_X)(Y_i - \hat{\mu}_Y)$$

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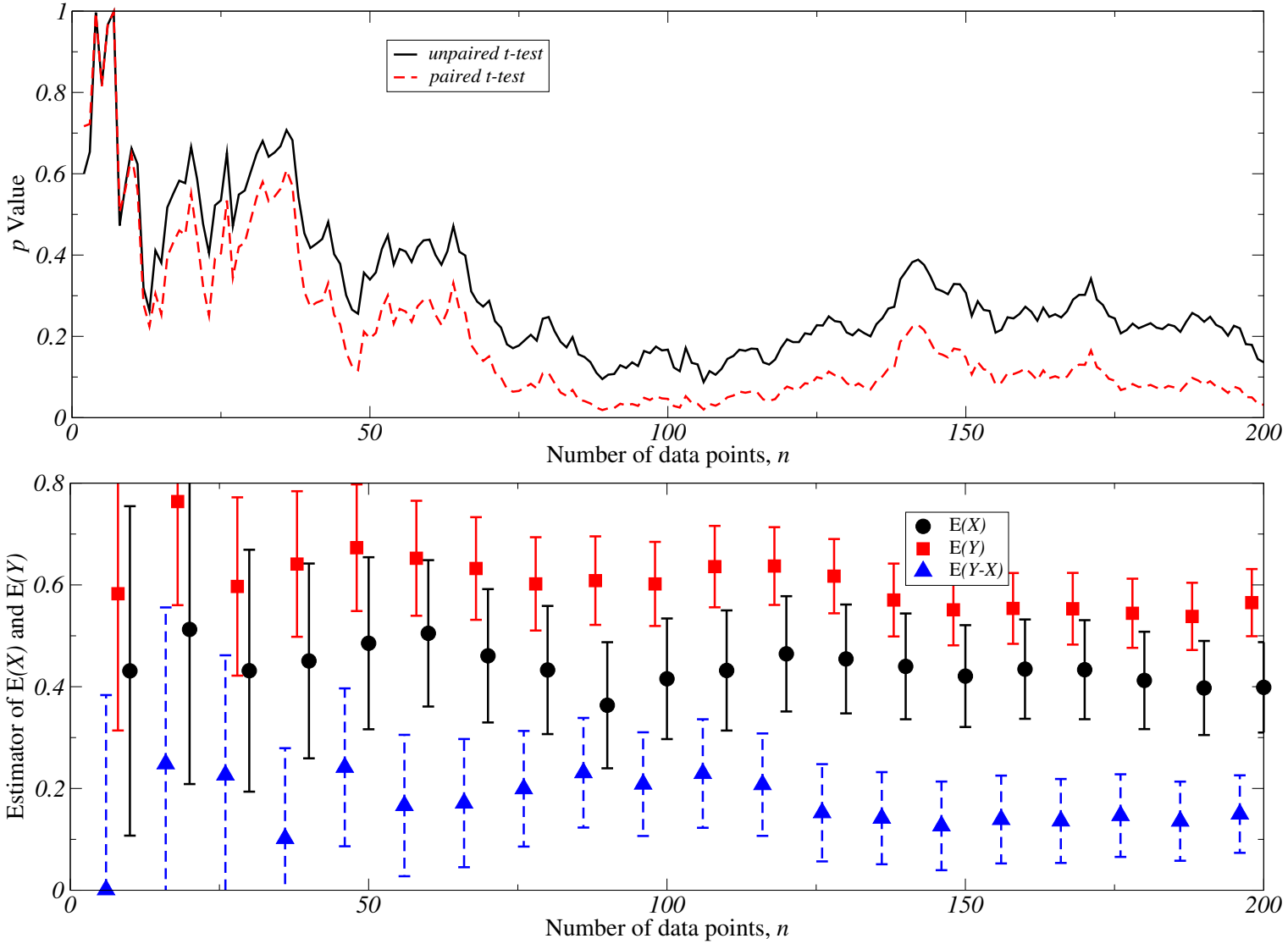
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Paired T-Test Example

$$X = Z + X_1 \qquad Y = Z + Y_1$$

$$Z \sim \mathcal{N}(0,0.8) \qquad X_1 \sim \mathcal{N}(0.4,0.2) \qquad Y_1 \sim \mathcal{N}(0.6,0.2)$$



Cheating Significance Tests

- You should only run your t-test once
- If you run multiple t-tests until you get one with $p < 0.05$ then you have cheated
- With a p value of 0.05 you expect to find the observed t value given the null hypothesis once every 20 trials

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Computing Significance Ab-Initio

- The t-test results follow from a beautiful, but difficult calculation
- However, you can simulate the null-hypothesis multiple times and measure empirically the number of times the t-value exceeds the observed t-values
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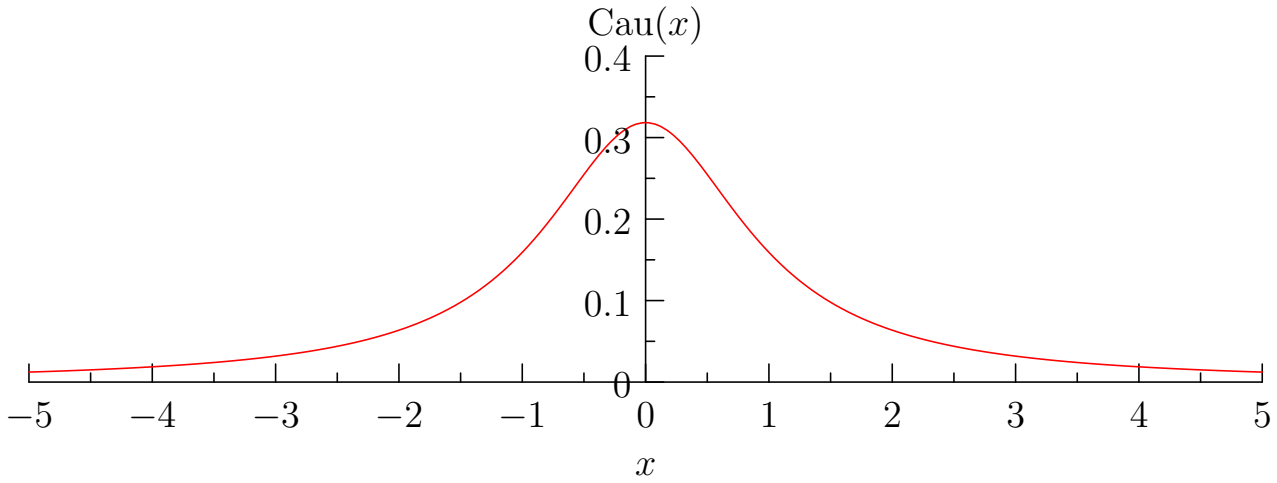
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- Just occasionally you come across data that has long (thick) tails

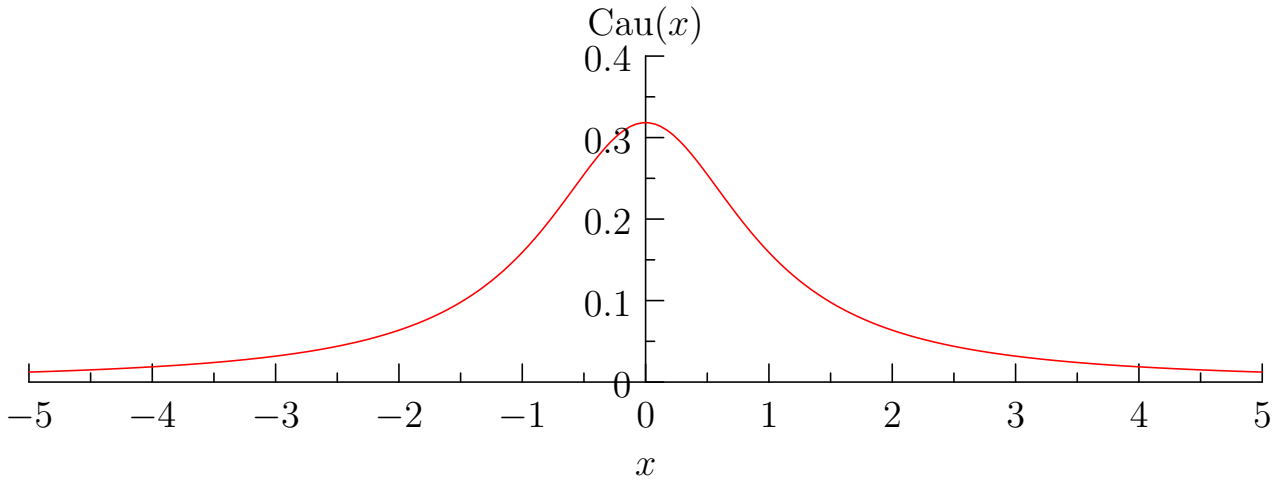


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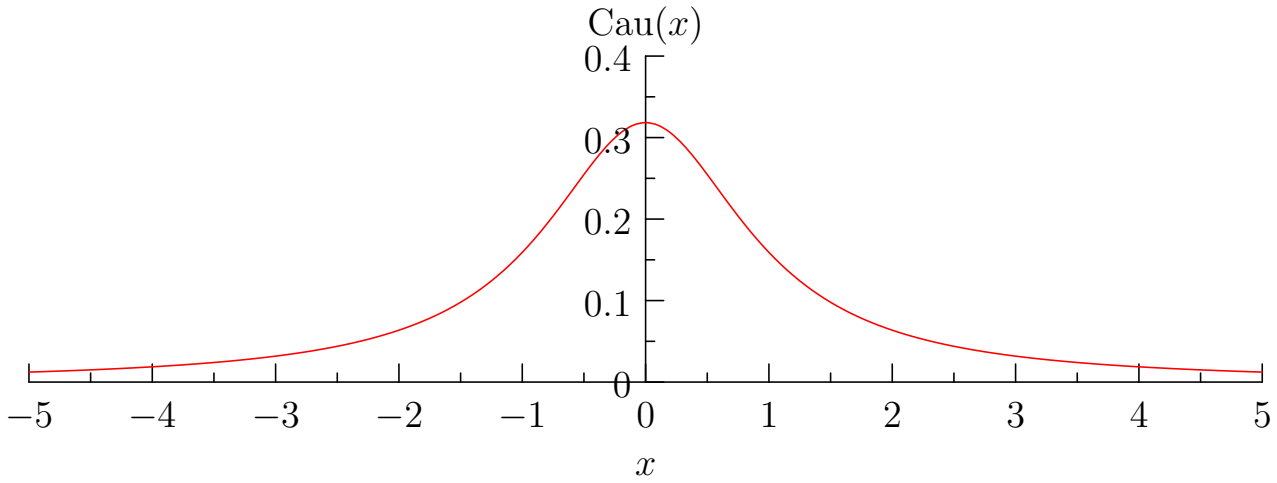


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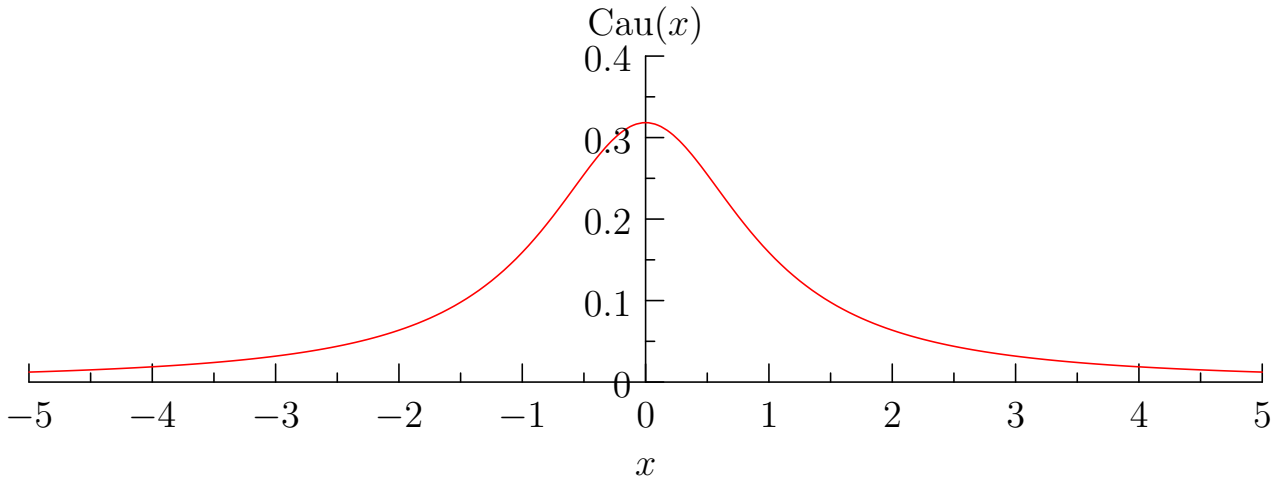


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